

NON-VALVULAR ATRIAL FIBRILLATION

Key Family Physicians: Dr Chen Jiawei / Dr Kwan Yew Seng

Specialist Advisor: Dr Lim Toon Wei (Senior Consultant, Department of Cardiology, NUHCS)

Acknowledgement for their inputs: NUHSP: Mr Marvin Sim

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INTRODUCTION

The estimated incidence of atrial fibrillation worldwide for men and women is 20.9 million and 12.6 million respectively, with a higher incidence in developed countries. Atrial fibrillation is independently associated with a two-fold increased all-cause mortality risk in women and 1.5-fold increase in men. Atrial fibrillation increases the risk of stroke by 3 to 5 times. In Singapore, 17% of strokes occurred in patients with atrial fibrillation. Non-valvular atrial fibrillation forms the majority of atrial fibrillation cases seen in NUP and is the focus of this guideline. Atrial Fibrillation associated with significant mitral stenosis or mechanical heart valves is associated with higher stroke risk and is not included.

Stages of Atrial Fibrillation

Stage	Description	Explanation
1	At risk of atrial fibrillation	Modifiable risk factors: obesity, lack of fitness, hypertension, sleep apnoea, excessive alcohol consumption, diabetes mellitus Non-modifiable risk factors: genetic factors (e.g., variants in <i>TTN</i> , <i>MYH7</i> , <i>MYH6</i> , <i>LMNA</i> , and <i>KCNQ1</i>), male sex, old age
2	Pre-atrial fibrillation	Structural or electrical conditions that can lead to atrial fibrillation (e.g., atrial enlargement, frequent atrial ectopy, short bursts of atrial tachycardia, atrial flutter, heart failure, valve disease, coronary artery disease, hypertrophic cardiomyopathy, neuromuscular disorders, thyroid disease)
3A	Paroxysmal atrial fibrillation	Intermittent and terminating within 7 days of onset
3B	Persistent atrial fibrillation	Continuous and lasting longer than 7 days
3C	Long-standing persistent atrial fibrillation	Continuous and lasting longer than 12 months
3D	Successful atrial fibrillation ablation	Freedom from atrial fibrillation after ablation

Stage	Description	Explanation
4	Permanent atrial fibrillation	Not pursuing further attempts at rhythm control

The 3 pillars of AF prevention and treatment are:

- lifestyle and risk-factor modification
- stroke prevention and
- symptom management

Primary Prevention of Atrial Fibrillation

Maintain or achieve a healthy weight
Engage in physical activity
Moderate alcohol consumption or abstain; avoid binge drinking
Stop smoking
Control hypertension
Control hyperglycaemia in diabetes

Secondary Prevention of Atrial Fibrillation

(for those who already have AF)

Lose weight if overweight or obese, i.e., body mass index > 27 kg/m ²
Start a standardized exercise program
Stop smoking

Minimize alcohol consumption or abstain entirely

Optimally control comorbidities including hypertension and diabetes

APPROACH

History

- Asymptomatic
- Palpitations
- Chest Pain or discomfort
- Shortness of Breath
- Giddiness, Syncope
- Decreased effort tolerance, Fatigue
- Anxiety
- Transient Ischemic Attack / Stroke
- Evaluation of associated co-morbidities:
 - Genetic Predisposition
 - Older Age (biggest risk factor for AF)
 - Hypertension
 - Heart Failure
 - Valvular Heart Disease
 - Myocardial Infarction
 - Thyroid Dysfunction
 - Obesity
 - Diabetes Mellitus
 - Chronic Obstructive Pulmonary Disease
 - Obstructive Sleep Apnoea
 - Chronic Kidney Disease
 - Smoking
 - Alcohol Consumption

Physical Examination

- Blood Pressure, Heart Rate
- Cardiovascular Examination: presence of cardiac murmurs, features of cardiac failure
- Neurological Examination
- Thyroid Dysfunction

Basic Investigations (for all patients)

- Electrocardiogram (ECG)
 - Absence of P waves, irregular undulating baseline, irregularly irregular R-R intervals
- Full blood count (FBC)
- Serum electrolytes
- Thyroid function test (TFT)
- Liver function test (LFT)
- Coagulation profile
- Transthoracic echocardiogram

Additional Investigations (for selected patients)

- Ambulatory ECG
- Transoesophageal echocardiogram
- Coronary angiography or cardiac stress testing

PRINCIPLES OF MANAGEMENT

In the acute setting, an initial assessment for the need for emergent care is performed. Red flags which warrant urgent referral to the Emergency Department include:

- Haemodynamically unstable
- Features of myocardial ischemia
- Features of congestive cardiac failure
- New onset focal neurological deficit
- Severe symptoms

1. Rate Control Therapy

Target a resting heart rate of < 110bpm.

Rate control drugs available in NUP

- Beta-blockers (e.g., bisoprolol, carvedilol, propranolol, metoprolol, atenolol)
 - First-line option unless contraindicated (severe asthma)
- Non-dihydropyridine calcium-channel blockers (e.g., diltiazem, verapamil)
 - When beta-blockers are not tolerated or are contraindicated
- Digoxin
 - In patients with heart failure with reduced ejection fraction
 - Has a narrow therapeutic window. Kidney function and electrolytes should be monitored in patients on digoxin
- Amiodarone

2. Monitoring of Patients on Long-Term Amiodarone

- Long-term Amiodarone use is associated with thyroid dysfunction, lung toxicity, liver toxicity, ophthalmologic and cardiovascular adverse effects
- Patients should be monitored with 6 monthly liver function test, thyroid function test and ECG
- Direct access referral to SOC is indicated for the following:
 - Referral to Endocrinology for abnormal thyroid function test (hypo- or hyperthyroidism)
 - Referral to Respiratory medicine for pneumonitis on CXR in patients with new onset or worsening cough or dyspnoea
 - Referral to Cardiology for transaminitis (> 3x ULN) or abnormal ECG (other significant arrhythmia, bradycardia < 50/min, QTc interval > 600ms)
- Avoid concomitant use of other drugs that prolong QTc interval

3. Rhythm Control Therapy

Sinus rhythm should be achieved early in the disease course with antiarrhythmic drugs or with catheter ablation to decrease the burden of atrial fibrillation (frequency and duration). Rhythm control in general is recommended to reduce the risk of progression and the risk of dementia or worsening cardiac structural abnormalities, and in patients with the following specific conditions:

- Reduced left ventricular function and persistent or high burden of atrial fibrillation
- Symptomatic atrial fibrillation
- Recently diagnosed symptomatic AF (< 1 year)
- Atrial fibrillation and heart failure

4. Stroke Prevention

- Prophylaxis against thromboembolism is the cornerstone of therapy for atrial fibrillation and should be considered regardless of whether a rate control or rhythm control therapy is chosen.
- Paroxysmal atrial fibrillation should be treated as for patients with persistent or longstanding persistent atrial fibrillation.
- The decision to initiate anti-coagulants depends on the individual patient's risk as determined by the CHA₂DS₂-VASc score.
- Anti-coagulation with DOACs is preferred over warfarin in patients without contraindication to DOACs as DOACs have a better safety profile and fewer food/drug interactions. As apixaban is SDL2 listed and rivaroxaban is MAF listed, drug differential cost between DOAC and warfarin is now limited. Patients would also benefit from cost savings through reduced clinic visits and INR testing.
- Antiplatelet agents are no longer recommended for stroke prevention and are only considered when anticoagulation is contraindicated in patients with high stroke risk. In patients with a previous history of stroke or TIA but who decline anticoagulation, it is reasonable to offer antiplatelet agents as secondary prevention for stroke.
- Concomitant treatment with anti-platelets leads to higher risk of bleeding and should be reserved for those with a clear indication for antiplatelet therapy on top of anticoagulation.

CHA₂DS₂-VASc Score

Condition		Points
C	Congestive Heart Failure	1
H	Hypertension (or treated hypertension)	1
A₂	Age ≥ 75 years	2
D	Diabetes	1
S₂	Prior stroke or TIA	2
V	Vascular disease	1
A	Age 65 to 74	1
Sc	Sex category (female)	1

Score	Adjusted Stroke Risk (%/yr)
0	0
1	1.3
2	2.2
3	3.2
4	4.0
5	6.7
6	9.8
7	9.6
8	6.7
9	15.2

Modified CHA₂DS₂-VASc Score

In the absence of other risk factors, female gender alone may not increase stroke risk. Local guidelines thus recommend the use of the modified CHA₂DS₂-VASc Score (mCHA₂DS₂-VASc), whereby gender does not contribute to the score.

Thus, the indication for the use oral anti-coagulants in patients with atrial fibrillation is as summarised:

mCHA ₂ DS ₂ -VASc = 0	No anticoagulation or antiplatelet recommended
mCHA ₂ DS ₂ -VASc = 1	Anticoagulation with Apixaban should be considered
mCHA ₂ DS ₂ -VASc ≥ 2	Anticoagulation recommended with DOAC (preferred option) or Warfarin

Considerations when mCHA₂DS₂VASc = 1

When mCHA₂DS₂VASc = 1, the decision to start OAC should consider patient-specific factors such as age or underlying conditions. Stroke risk is higher for these patient groups:

- Age 65 – 74 years
- Heart failure and age ≥ 35 years
- Hypertension and age ≥ 50 years
- Diabetes mellitus and age ≥ 50 years
- Vascular diseases and age ≥ 55 years

HAS-BLED RISK SCORE

	Condition	Points
H	Hypertension (systolic blood pressure > 160mmHg)	1
A	Abnormal renal and liver function (1 point each)	1 or 2
S	Stroke	1
B	Bleeding tendency or predisposition	1
L	Labile INRs (for patients taking warfarin)	1
E	Elderly (Age > 65)	1
D	Drugs (concomitant aspirin or NSAIDS) or alcohol abuse (1 point each)	1 or 2

Score	Bleeds per 100 patient-years
0	1.13
1	1.02
2	1.88
3	3.74
4	8.70
5 to 9	Insufficient data

- The HAS-BLED risk score poorly predicts bleeding events but is a useful tool for identifying modifiable risks factors of bleeding.
- A high HAS-BLED score (≥ 3) indicates high bleeding risk but is not a contraindication for anticoagulation. These patients should be monitored more closely and every effort should be made to correct modifiable risk factors as much as possible to reduce bleeding risk.

Consider aspirin or clopidogrel only when anti-coagulation is contraindicated in patients with $mCHA_2DS_2-VASc \geq 2$, especially in those with a history of ischemic stroke or transient ischemic attack.

ANTICOAGULATION PROTOCOL IN NUP

Direct Oral Anticoagulants (DOACs) / Non-Vitamin K Antagonist Anticoagulants (NOACs), Rivaroxaban and Apixaban, and Warfarin are available in the NUP formulary. Initiation of anticoagulation is usually done inpatient or at the Specialist Outpatient clinics. In NUP, initiation of and titration of oral anticoagulants should be supervised by a Family Physician or designated Senior Physician.

DIRECT ORAL ANTICOAGULANTS (DOACs)

Rivaroxaban and Apixaban are factor Xa inhibitors that are available in the NUP formulary.

DOACs are as effective as warfarin in reducing atrial fibrillation related strokes and systemic embolisms in patients with i) non-valvular heart disease and ii) aortic stenosis, aortic or mitral regurgitation. There is limited evidence in patients with bioprosthetic heart valves, tricuspid regurgitation or mild mitral stenosis, but this is generally considered safe.

The use of DOAC in mechanical heart valves or moderate to severe mitral stenosis is not recommended. These patients should be treated with warfarin.

DOACs are associated with fewer intracranial haemorrhages compared to warfarin but may be associated with increased gastrointestinal bleeding except for Apixaban which has similar GI bleed risk. Limited experience with reversal agents may make DOACs less preferable in patients with high bleeding risks.

CONTRAINDICATIONS TO DOACs

- Hypersensitivity
- Clinically significant active bleeding
- Hepatic disease with coagulopathy and clinically relevant bleeding risk
- Pregnancy and lactation
- Lesion or condition considered to be a significant risk of major bleeding (recent GI ulceration, pregnancy, malignant neoplasm at high risk of bleed *etc.*)
- Concomitant treatment with any other anticoagulant agent
- History of intracranial bleed
- Renal impairment with creatinine clearance (Cockcroft-Gault) < 15 ml/min for Rivaroxaban and Apixaban
- Mechanical heart valves
- Moderate – Severe Mitral Stenosis

DOAC PROTOCOL IN NUP

In NUP, DOACs may be initiated in suitable patients with newly diagnosed atrial fibrillation and high stroke risk by a family physician or clinic designated senior physician, while awaiting echocardiogram and cardiology review. Patients identified for initiation of anticoagulation should be counselled regarding treatment options including DOACs and warfarin. Adverse reactions related to increased risk of bleeding including possible serious, potentially life-threatening intracranial or gastrointestinal bleeding must be explained.

For patients on antiplatelet therapy who develop new AF, management depends on the indication for antiplatelet therapy. For primary cardiovascular prevention, switch to anticoagulation monotherapy is recommended. For PCI with stable ischemic heart disease or acute coronary syndrome, use of oral anticoagulant plus a P2Y12 inhibitor for no more than 12 months is recommended, followed by oral anticoagulation alone. For patients with cerebrovascular disease without carotid stenting, oral anticoagulation monotherapy is recommended. For patients with carotid stenting or peripheral artery disease, a short course of anticoagulation plus P2Y12 inhibitor may be recommended, followed by oral anticoagulation alone.

Laboratory investigations for newly diagnosed atrial fibrillation prior to initiation of DOAC

1. Full Blood Count
2. Liver Function Test
3. Renal Panel
4. PT / APTT
5. Thyroid Function Test

Patients who are initiated on DOAC in polyclinic (while awaiting an echocardiography, or cardiology clinic review) should fulfil the following criteria:

1. No contraindications to DOAC
2. No features of mitral stenosis on clinical examination
3. Normal FBC, renal panel, LFT, PT/APTT
4. Any of the following features of high stroke risk:
 - a) History of stroke / TIA (regardless of other factors)
 - b) Age $\geq$ 75 years old
 - c) modified CHA₂DS₂VASc score of $\geq$ 2

Dose Adjustments for DOAC for Non-Valvular Atrial Fibrillation

1. Apixaban
 - 5mg BD
 - 2.5mg BD if patients fulfils either ABC Criteria or CrCl criteria

ABC Criteria (If patients have ≥ 2 of the following)	CrCl Criteria*
• Age ≥ 80 years • Body Weight ≤ 60kg • Serum Creatinine ≥ 133umol/L	15 – 29ml/min *Based on actual body weight

- CrCl < 15ml/min: Avoid use

2. Rivaroxaban
 - CrCl > 50 ml/min: Rivaroxaban 20mg OD
 - CrCl 15 - 50 ml/min: Rivaroxaban 15mg OD
 - CrCl < 15 ml/min: Avoid use

Recommended Follow Up After Initiation of DOAC

1. Review patient 1-3 months after initiation with Full Blood Count and Renal Panel.
2. Monitoring to include symptoms of stroke, transient ischaemic attack, thromboembolism and bleeding events
3. Consider 6 monthly reviews subsequently (more frequent if any concurrent condition that may impact use)
4. Monitor age, body weight, frailty, fall risk and changes in concomitant medications at every visit
5. Monitor Full Blood Count and Renal Panel, Liver Function Test annually and as clinically indicated.

DRUG COST

Drug	Subsidy Status	Cost per tablet	Cost per week (Adult)*	Cost per week (Elderly)*
Warfarin	SDL 1	\$0.20	\$1.40 (capped)	\$0.70 (capped)
Apixaban	SDL 2	\$0.77	\$5.39	\$2.70
Rivaroxaban**	SDL 2***	\$1.54	\$5.39	\$2.70

* Cost does not include additional MG / PG subsidy

** Cost does not take into account MAF subsidy which is determined through means testing

*** Change in class will take effect 1st April 2026

Switching of Warfarin to DOACs

Patients with existing non-valvular AF on warfarin may be offered conversion to a DOAC if they are:

- 1) Unable to maintain therapeutic INR
- 2) have difficulty attending clinic frequently for INR monitoring
- 3) have problematic drug interactions, or
- 4) find the dietary restrictions associated with warfarin therapy interfere excessively with a healthy, balanced diet

INR <2.5: Stop Warfarin and start DOAC on the same day.

INR 2.5-3: Stop warfarin and start DOAC the next day.

INR >3: Repeat INR and start DOAC as per the above recommendations once INR has fallen below 3.

When switching from warfarin to DOAC, the same protocol for the initiation of DOACs and subsequent follow-up should be followed as per the previous section.

WARFARIN PROTOCOL IN NUP

Medical Officers (MOs) and Resident Physicians (RPs) may provide follow up care for patients on Warfarin, and repeat prescriptions where INRs are within range and no titration of Warfarin dose is required. Where titration of Warfarin is indicated, MOs / RPs will need to seek approval of proposed warfarin dose adjustment by FPs or designated RPs. Documentation of dosage change should be made clearly in the patient's electronic notes and e-prescription module. Pharmacy staff will ensure that any change in Warfarin dosage or initiation of Warfarin is countersigned before they dispense the medication. Where alteration of dosage is involved, doctors should:

- a) Observe for bleeding: haematuria, melena, gingival bleeding, excessive bleeding from cuts, epistaxis, bruising, dizziness, hypotension, weakness
- b) Observe for thromboembolic event: DVT, pulmonary embolism, CVA, AMI
- c) Exclude any dietary or drug interaction that may affect patient's INR
- d) Check haemoglobin and haematocrit annually

For polyclinics with Anti-Coagulation Clinic (ACC) service run by Pharmacists, the FP will work in collaboration with the pharmacist in the shared care follow up of these patients.

During follow-up at the polyclinics, patients on Warfarin should be offered Warfarin education by Doctors or Pharmacists, which may consist of education about Warfarin and issuing of health education materials, e.g., pamphlets on Warfarin General education on Warfarin would include understanding Warfarin dosage, assessment of medication compliance, haemorrhagic side effects related to Warfarin and potential for drug or food interaction. Patients may require additional Warfarin counselling with regards to a review of dietary habits, review of possible drug or food interactions. Such patients may be referred to pharmacists for counselling on an ad hoc basis. The doctor or pharmacist should state the date of Warfarin education conducted in the electronic notes.

Patients should be educated that when they undergo any form of surgery or dental procedure, they must inform the doctor of their Warfarin therapy so that appropriate steps are taken to ensure safety during the procedures. The aim is to avoid excessive post-procedural bleeding.

Initiation / Re-Initiation of Warfarin

Warfarin can be initiated or reinstated in the polyclinic after it has been discontinued temporarily (e.g., after a procedure)

- Check baseline INR before initiation of warfarin. Consider evaluation for coagulopathy before initiating warfarin if INR > 1.2
- Re-initiation in polyclinic should not be attempted for cases that require low molecular weight heparin during procedure. Cases requiring rapid anticoagulation with concurrent heparin / low molecular weight heparin should be referred to hospitals for management
- For initiation, a fixed dose of 2 - 5 mg/day (2 - 3 mg for Chinese / Malays and 4 - 5 mg for Indians) is recommended
- For re-initiation:
 - Exclude development of any new contraindications to warfarin
 - Restart patient on last known warfarin dose that maintained patient at therapeutic range
- Check INR on day 3 after initiation / re-initiation and every 1 - 2 days till 2 consecutive readings of target INR have been achieved
- Generally, doses will be adjusted by 0.5 mg to 1 mg
- Steady state is expected at least after 5 days
- Thereafter check INR weekly for the first month, then 4 weekly and finally 8 to 12 weekly once INR readings are stable
- Adjust INR accordingly via the intranet warfarin calculator if not at therapeutic range after 1 - 2 weeks post initiation

INR Targets When Using Warfarin

- Target INR for most patients will be within the range of 2.0-3.0.
- In elderly patients > 75 years of age, or in those deemed to be at higher bleeding risk, a lower INR target of 1.6 - 2.5 may be chosen instead, balancing the risks and benefits of anticoagulation.
- Patients with higher thrombotic risk may have a target INR of 2.5-3.5.

Warfarin Initiation Guide (Please note this is a guide. Physician to exercise discretion, based on clinical experience)

Day	INR	Dose (mg) for Chinese / Malay	Dose (mg) for Indian
1	Baseline	3	5
2	-	3	5
3	< 1.2	3	5
	1.2 - < 1.5	3	5
	1.5 - < 2.0	3	5
	2.0 - < 3.0	2	3
	≥ 3.0	Nil	Nil
4	< 1.3	5	8
	1.3 - < 1.5	4	6.5
	1.5 - < 1.7	3	5
	1.7 - < 2.0	2.5	4
	2.0 - < 2.5	2.0	3
	2.5 - < 3.0	1.5	2.5
	3.0 - < 3.5	Omit for 1 day, then 1 mg	Omit for 1 day, then 2 mg
	3.5 - < 4.0	Omit for 1 day, then 1 mg	Omit for 1 day, then 2 mg
	≥ 4.0	Omit for 2 days, then 0.5 mg	Omit for 2 days, then 1 mg

For example, a 55 years old Chinese male initiating on warfarin:

Day	0	1	2	3	4	5	6	7	8	9	10	11 to 14
Warfarin Dose (mg)		3.0	3.0	3.0	2.0	2.0	2.5	2.5	2.5	2.5	2.5	2.5
INR	1.0	-	-	2.1	-	1.9	-	2.2	-	-	2.1	Check INR Day 14

CONTRAINDICATIONS TO WARFARIN

Patients should not continue with warfarin if they present with a history of the following:

- Aneurysm (cerebral or dissecting)
- Active bleeding disorder or unexplained anaemia
- Cerebral vascular haemorrhage (confirmed or suspected), unless cleared by neurologist or neurosurgeon.
- Blood dyscrasias associated with haemorrhage or thrombocytopenia
- Severe uncontrolled hypertension
- Recent (2 - 3 weeks) trauma (especially to the CNS)
- Neurosurgery unless cleared by neurosurgeon
- Ulceration or active lesions of the GIT, respiratory or urinary tracts
- Severe vasculitis
- Pregnancy (1st trimester and before delivery)

Other factors making warfarin therapy unsuitable are:

- Alcohol abuse
- At high risk of falls
- Poor family support / inability to return for regular INR monitoring
- Non-compliance with medication
- Mentally unstable patient

Caution should be exercised in the following conditions which are associated with increased sensitivity to warfarin:

- Age > 75 years
- Clinical congestive cardiac failure
- Drug interactions, e.g., drugs which inhibit warfarin metabolism.
- Elevated baseline INR
- Hypermetabolic states
- Liver or renal impairment
- Malnutrition / low vitamin K intake
- Thyrotoxicosis

Drugs with Major Interactions with Warfarin (not an exhaustive list)

↑ Increase Effect of Warfarin		↓ Decrease Effect of Warfarin
<u>Anti-platelets and Anticoagulants</u> Aspirin, Clopidogrel, Dipyridamole, Ticlopidine Apixaban, Dabigatran, Rivaroxaban <u>Endocrine</u> Levothyroxine <u>Androgens, Estrogens, Progestins</u> <u>Sulphonylureas</u> <u>Cardiovascular</u> Amiodarone Diltiazem, Verapamil Propranolol Simvastatin, Lovastatin Fenofibrate, Gemfibrozil <u>Psychiatric</u> <u>Selective Serotonin Reuptake Inhibitors</u> (e.g., Escitalopram, Fluoxetine, Fluvoxamine, Sertraline) <u>Tricyclic antidepressants</u> (e.g., Amitriptyline, Doxepin) Mirtazapine Venlafaxine Quetiapine <u>Anticonvulsants</u> Valproic acid Phenytoin	<u>Malignant disease and Immunosuppression</u> Methotrexate Tamoxifen <u>Musculoskeletal</u> Allopurinol <u>Corticosteroid</u> Prednisolone <u>Gastrointestinal</u> Cimetidine, Ranitidine Omeprazole	<u>Anticonvulsants</u> Carbamazepine Phenobarbital Phenytoin <u>Endocrine</u> <u>Anti-thyroid agents</u> (e.g., Carbimazole, Propylthiouracil) <u>Cardiovascular</u> Cholestyramine <u>Malignant disease and Immunosuppression</u> Azathioprine Sulphasalazine <u>Antibiotics and Antifungals</u> Rifampicin
	<u>ACUTE MEDICATIONS</u> <u>Analgesics</u> NSAIDs, COX-II Selective inhibitors Paracetamol Tramadol <u>Antibiotics and Antifungals</u> Azithromycin Clarithromycin Ciprofloxacin Erythromycin Metronidazole Trimethoprim-Sulfamethoxazole Itraconazole	

WARFARIN-FOOD INTERACTIONS

Foods Rich in Vitamin K Decrease Effect of Warfarin

Green leafy vegetables (e.g., spinach, broccoli, lettuce, Brussels sprouts), certain legumes, some vegetable oils (e.g., soybean oil) animal livers, some fermented foods (e.g., cheese), green tea. Chronic alcohol intake can increase metabolism of oral anticoagulants.

Foods with Anti-Platelet Effect

Garlic, foods containing salicylates (fruits, vegetables, spices, teas, certain flavoured candies)

Others

Avocado (decrease effect of warfarin), Vitamin E (potentiate effect of warfarin), dietary supplements [arnica, bilberry, butchers broom, cat's claw, dong quai, feverfew, forskolin, garlic, ginger, ginkgo, horse chestnut, inositol hexaphosphate, licorice, melilot (sweet clover), pau d'arco, red clover, St John's wort, sweet woodruff, turmeric, willow bark, wheat grass, alcohol (continuous heavy drinking stimulates hepatic enzymes, increasing metabolism of warfarin, decrease effect of warfarin)]

Monitoring Using INR

Repeat INR every 1 to 2 weeks for every dose adjustment.

When target INR is achieved, next INR may be checked after 4 weeks and subsequently 8 to 12 weekly.

Management of Supratherapeutic INR:

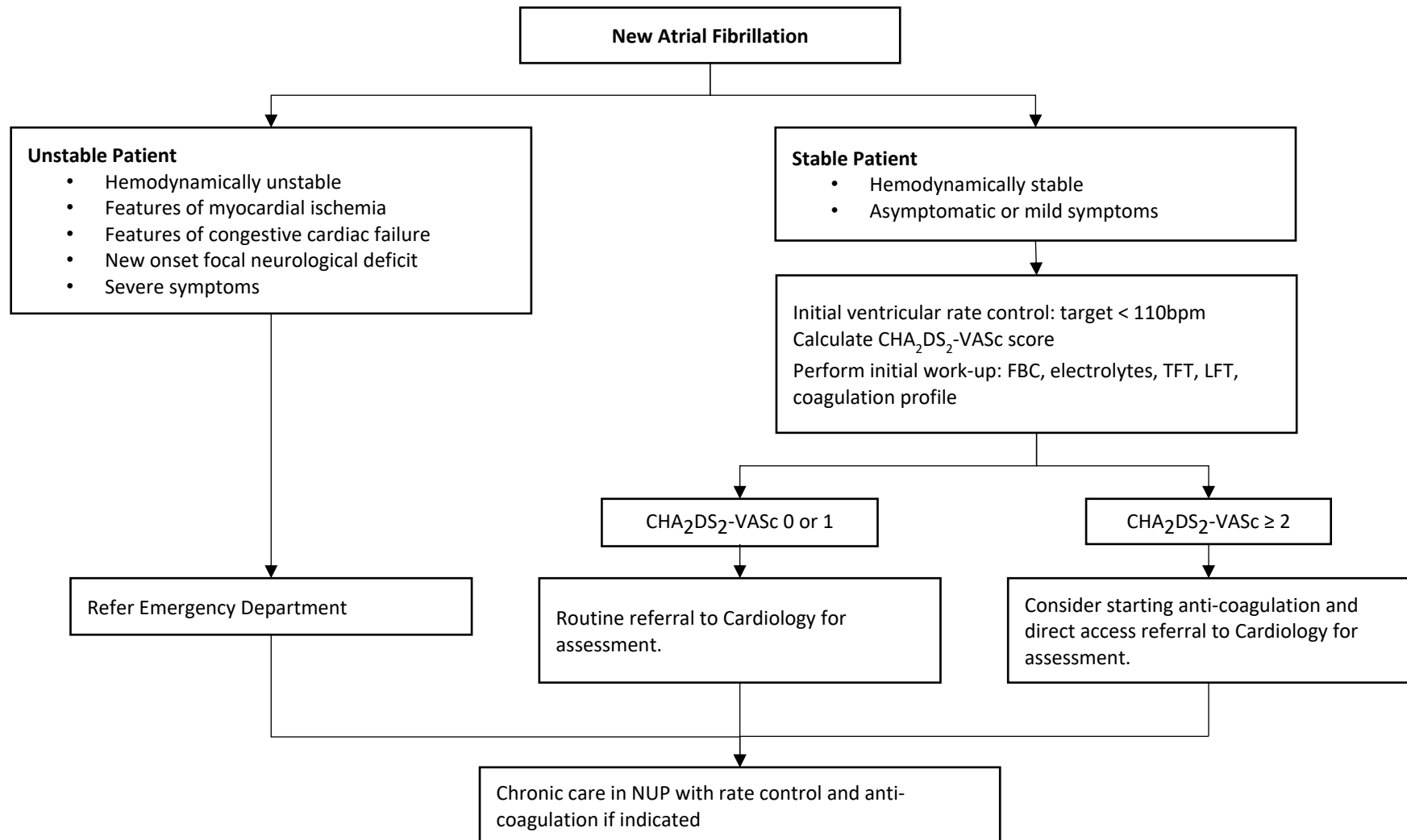
INR	Management
Greater than therapeutic range but < 4.5	Decrease or withhold dosage Monitor INR more frequently and restart warfarin at a lower dose when INR is within therapeutic range.
4.5 – 9.0	Withhold warfarin, consider referral to Emergency Department for oral Vitamin K If managed in primary care, recheck INR within 24-28 hours. If within therapeutic range, resume warfarin at a lower dose.
> 9.0	Referral to Emergency Department for oral Vitamin K

SUMMARY OF ORAL ANTICOAGULANTS

	Warfarin	Rivaroxaban	Apixaban
Mechanism of Action	Vitamin K Antagonist	Oral anti-Xa inhibitor	Oral anti-Xa inhibitor
Metabolism	Major CYP2C9, CYP450, CYP1A2, CYP3A4 substrate	CYP3A3 substrate P-glycoprotein substrate	CYP3A3 substrate P-glycoprotein substrate
Tmax (hours)	4	3-4	3
Half-Life (hours)	36	5-13	9-14
Elimination	Hepatic (cytochrome P450)	Liver (66%) Renal (33%)	Renal (27%)
Drug interactions	Refer to page 11	Contraindicated: clinically significant increase in levels	
		Strong CYP3A4 inhibitors AND/OR P-gp inhibitors: systemic azole antifungals (except fluconazole), HIV protease inhibitors	
		Caution: avoid use/seek advice, may need dose change	
		Less potent P-gp and/or CYP3A4 inhibitors: amiodarone, erythromycin, clarithromycin, fluconazole, diltiazem, quinidine, tacrolimus, verapamil	
		Potent P-gp ± CYP3A3 inducers: carbamazepine, phenobarbital, phenytoin, rifampicin, St John's Wort	
		Antiplatelet agents / anticoagulants	
		Some antidepressants: SSRI or SNRIs	
Special Precautions	Ensure consistent Vitamin K intake in diet. Avoid supplements	Take with food to increase bioavailability	With or without food
Tube Feeding Considerations	Separate by 1 hour	Crush or suspend in 50ml of water and give via NG or gastric feeding tube. Dose should be immediately followed by enteral feeding. Suspension is stable in water for up to 4h.	Crush the tablet and mix the medicine in a syringe with 60ml of 5% dextrose in water. Give this mixture right away through the NG tube. Do not save for later use.
Laboratory Tests	PT/INR every 2-3 monthly and FBC at least annually	FBC, Renal Panel at least annually	

SUMMARY OF MANAGEMENT OF NEWLY DIAGNOSED ATRIAL FIBRILLATION

All cases of newly diagnosed atrial fibrillation are referred for further cardiologist assessment. The urgency of the referral will depend on the severity of symptoms, hemodynamic stability and stroke risk as determined by the CHA₂DS₂-VASc score. Suitable patients may be started on anti-coagulation for early stroke prevention while awaiting cardiologist assessment.



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